Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: P&D West Dairy

Physical address of dairy:

29631 Husman RD	Gustine	Merced	95322
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1930

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X070-X060-X043-XXXX

B. OPERATORS

Lopes, Paul

Operator name: Lopes, Paul

Telephone no.: Landline: ; Cellular: (209) 649-4967

9270 S Whitworth RD	Gustine	CA	95322
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Lopes, Paul

Legal owner name: Lopes, Paul

Telephone no.: Landline: ; Cellular: (209) 649-4967

9270 S Whitworth RD	Gustine	CA	95322
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	475	0	0	0	0	0
Maximum number	475	0	0	0	0	0
Average number	475	0	0	0	0	0
Avg live weight (lbs)	1,400	0	0	0		

Predominant milk cow breed: Holstein

Average milk production: 68 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 12,072.29 *tons per reporting period*

Total nitrogen from manure: 157,847.81 *lbs per reporting period*

After ammonia losses (30% loss applied): 110,493.47 *lbs per reporting period*

Total phosphorus from manure: 26,701.43 *lbs per reporting period*

Total potassium from manure: 90,342.73 *lbs per reporting period*

Total salt from manure: 223,653.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 12,480,000 *gallons*

Total nitrogen generated: 42,893.97 *lbs*

Total phosphorus generated: 8,188.25 *lbs*

Total potassium generated: 74,335.92 *lbs*

Total salt generated: 514,278.98 *lbs*

	12,480,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	12,480,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well	Ground water
CCID	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 50	57	57	2	process wastewater	X070-X060-X003-XXXX
Field 51	51	51	2	both	X070-X060-X042-XXXX X070-X060-X043-XXXX
Totals for areas that were used for application	108	108	4		
Totals for areas that were not used for application					
Land application area totals	108	108	4		

B. CROPS AND HARVESTS

Field 50

Field name: Field 50

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 57 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/18/2023	914.00 *ton*	As-is		79.0	5,860.00	1,320.00	18,900.00		13.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	16.04	187.93	42.33	606.13	895.72

04/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 57 Plant date: 04/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/26/2023	1,359.00 *ton*	As-is		70.9	3,840.00	1,400.00	6,810.00		8.57

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	23.84	183.11	66.76	324.73	1,189.18

Field 51

Field name: Field 51

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 51 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/18/2023	653.00 *ton*	As-is		65.6	5,060.00	1,270.00	18,900.00		9.47

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	12.80	129.58	32.52	483.99	834.22

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 51 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	991.00 *ton*	As-is		61.6	4,950.00	1,240.00	6,460.00		12.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	19.43	192.37	48.19	251.05	1,790.80

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 50 - 11/01/2022: Oats, silage-soft dough

Field name: Field 50

Crop: Oats, silage-soft dough — Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	147.05	20.49	237.17	1,688.68	2,160,000.00 *gal*
Ag Well	Ground water	4.23	0.00	0.00	319.39	4,320,000.00 *gal*
Application event totals		151.28	20.49	237.17	2,008.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.47	17.60	102.46	1,033.02	1,440,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	147.05	6,480,000.00 *gal*
Application event totals		71.47	17.60	102.46	1,180.07	

Field 50 - 04/28/2023: Corn, silage

Field name: Field 50

Crop: Corn, silage — Plant date: 04/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	257.33	11,340,000.00 *gal*
Application event totals		0.00	0.00	0.00	257.33	

Field 50 - 04/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	147.05	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	147.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.47	17.60	102.46	1,033.02	1,440,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	183.81	8,100,000.00 *gal*
Application event totals		71.47	17.60	102.46	1,216.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	183.81	8,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	183.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.99	15.94	177.09	1,222.76	1,440,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	147.05	6,480,000.00 *gal*
Application event totals		104.99	15.94	177.09	1,369.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	147.05	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	147.05	

Field 50 - 04/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.49	7.97	88.54	611.38	720,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	147.05	6,480,000.00 *gal*
Application event totals		52.49	7.97	88.54	758.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	147.05	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	147.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	110.29	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	110.29	

Field 51 - 11/01/2022: Oats, silage-soft dough

Field name: Field 51

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	109.56	15.27	176.72	1,258.23	1,440,000.00 *gal*
Ag Well	Ground water	3.94	0.00	0.00	297.47	3,600,000.00 *gal*
Application event totals		113.51	15.27	176.72	1,555.70	

Field 51 - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	83.18	14.51	159.75	1,008.47	1,440,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	123.26	4,860,000.00 *gal*
Application event totals		83.18	14.51	159.75	1,131.73	

Field 51 - 06/19/2023: Corn, silage

Field name: Field 51

Crop: Corn, silage

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	118.15	22.08	58.30	3,105.56	420.00 *ton*
Application event totals		118.15	22.08	58.30	3,105.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	212.28	8,370,000.00 *gal*
Application event totals		0.00	0.00	0.00	212.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	184.89	7,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	184.89	

Field 51 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.23	11.88	131.95	911.08	960,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	143.80	5,670,000.00 *gal*
Application event totals		78.23	11.88	131.95	1,054.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	191.74	7,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	191.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.99	14.96	99.08	329.87	720,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	143.80	5,670,000.00 *gal*
Application event totals		34.99	14.96	99.08	473.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	164.35	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	164.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.99	14.96	99.08	329.87	720,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	157.50	6,210,000.00 *gal*
Application event totals		34.99	14.96	99.08	487.37	

Field 51 - 06/19/2023: Corn, silage

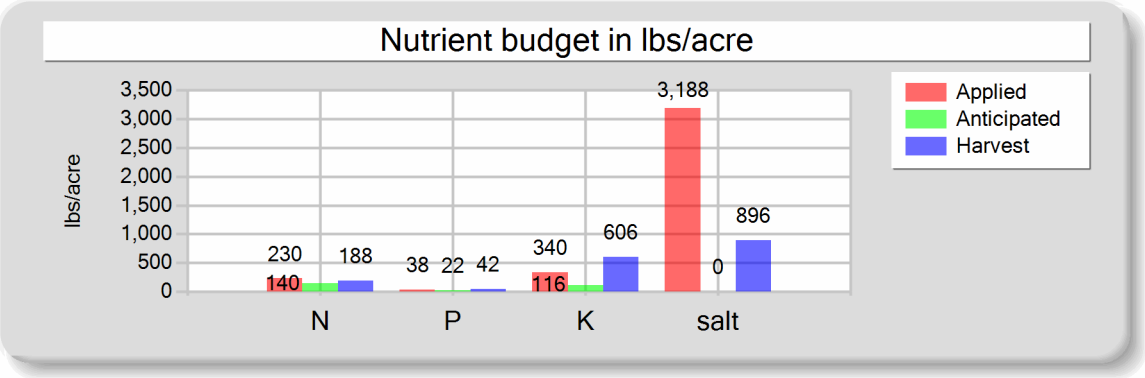
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	143.80	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	143.80	

B. NUTRIENT BUDGET

Field 50 - 11/01/2022: Oats, silage-soft dough

Field name: Field 50 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	218.52	38.10	339.63	2,721.70
Fresh water	4.23	0.00	0.00	466.44
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.75	38.10	339.63	3,188.14
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	187.93	42.33	606.13	895.72
Nutrient balance	41.82	-4.24	-266.49	2,292.42
Applied to removed ratio	1.22	0.90	0.56	3.56

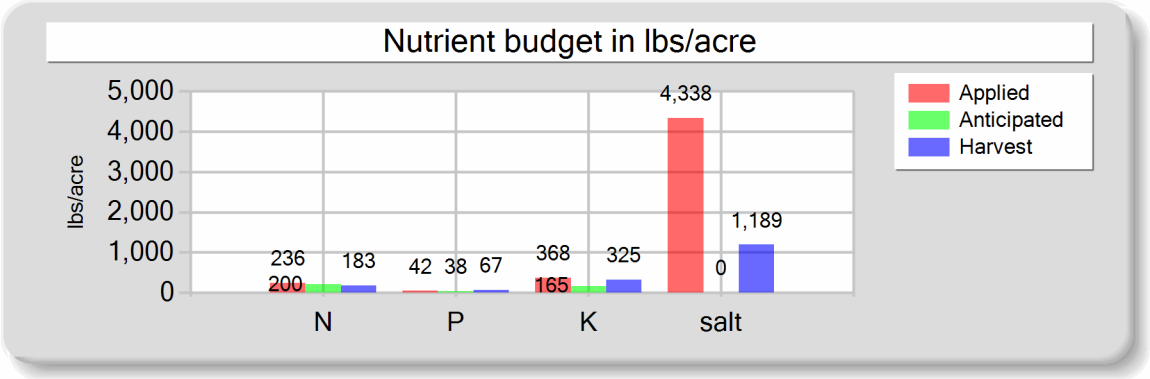
Fresh water applied	
10,800,000.00	*gallons*
397.73	*acre-inches*
6.98	*inches/acre*

Process wastewater applied	
3,600,000.00	*gallons*
132.58	*acre-inches*
2.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 50 - 04/28/2023: Corn, silage

Field name: Field 50 Crop: Corn, silage Plant date: 04/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.95	41.51	368.09	2,867.17
Fresh water	0.00	0.00	0.00	1,470.48
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.95	41.51	368.09	4,337.64
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	183.11	66.76	324.73	1,189.18
Nutrient balance	52.84	-25.25	43.36	3,148.46
Applied to removed ratio	1.29	0.62	1.13	3.65

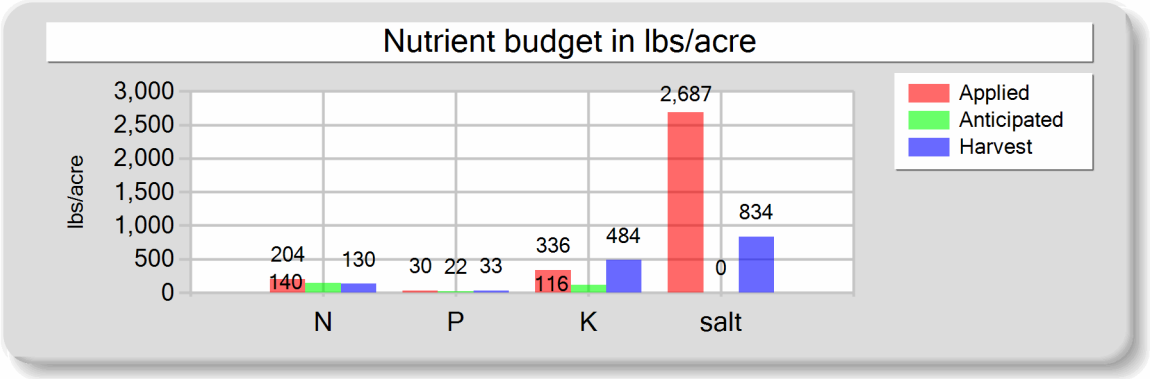
Fresh water applied	
64,800,000.00	*gallons*
2,386.36	*acre-inches*
41.87	*inches/acre*

Process wastewater applied	
3,600,000.00	*gallons*
132.58	*acre-inches*
2.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 51 - 11/01/2022: Oats, silage-soft dough

Field name: Field 51 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	192.74	29.78	336.47	2,266.70
Fresh water	3.94	0.00	0.00	420.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	203.68	29.78	336.47	2,687.43
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	129.58	32.52	483.99	834.22
Nutrient balance	74.11	-2.74	-147.52	1,853.21
Applied to removed ratio	1.57	0.92	0.70	3.22

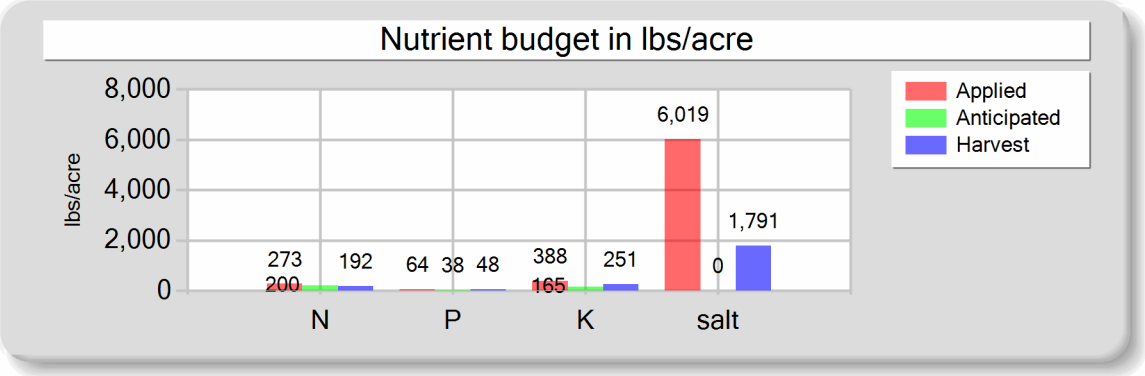
Fresh water applied	
8,460,000.00	*gallons*
311.55	*acre-inches*
6.11	*inches/acre*

Process wastewater applied	
2,880,000.00	*gallons*
106.06	*acre-inches*
2.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 51 - 06/19/2023: Corn, silage

Field name: Field 51 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	118.15	22.08	58.30	3,105.56
Process wastewater	148.21	41.80	330.11	1,570.82
Fresh water	0.00	0.00	0.00	1,342.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	273.36	63.88	388.41	6,018.56
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	192.37	48.19	251.05	1,790.80
Nutrient balance	80.99	15.69	137.36	4,227.76
Applied to removed ratio	1.42	1.33	1.55	3.36

Fresh water applied	
52,920,000.00	*gallons*
1,948.86	*acre-inches*
38.21	*inches/acre*

Process wastewater applied	
2,400,000.00	*gallons*
88.38	*acre-inches*
1.73	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall Solid Manure

Sample and source description: Fall Solid Manure
Sample date: 10/21/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,000.00	4,700.00	5,720.00	0.00	0.00	0.00	0.00	0.00		0.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Spring Solid Manure

Sample and source description: Spring Solid Manure
Sample date: 05/03/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 60.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,300.00	3,420.00	9,030.00	14,400.00	6,850.00	3,370.00	3,300.00	1,020.00		48.10
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Fall Solid Manure

Sample and source description: Fall Solid Manure
Sample date: 10/05/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 40.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	3,790.00	10,600.00	0.00	0.00	0.00	0.00	0.00		70.70
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 12/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	465.00	355.00	0.00	0.00	64.80	750.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6,960.00	5,340
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 12/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	465.00	355.00	0.00	0.00	64.80	750.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6,960.00	5,340
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 03/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	339.00	221.00	0.00	0.00	83.50	486.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5,190.00	4,900
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	498.00	378.00	0.00	0.00	75.60	840.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	7,120.00	5,800
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	5.00	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 10/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	297.00	235.00	0.00	0.00	127.00	841.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4,810.00	2,800
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 12/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	633.00	406.00	0.00	0.00	73.00	874.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	7,190.00	8,720
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

C. FRESH WATER ANALYSES

Ag Well

Ag Well

Sample description: Ag Well

Sample date: 01/03/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.69	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,100.00	505
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

CCID

CCID

CCID

Sample description: CCID

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								228.00	155
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 50 - 11/01/2022: Oats, silage-soft dough

Oats

Sample and source description: Oats

Sample date: 04/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 79.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,860.00	1,320.00	18,900.00		13.30
DL	100.00	100.00	100.00		0.01

Field 50 - 04/28/2023: Corn, silage

Field 50 - 04/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,840.00	1,400.00	6,810.00		8.57
DL	100.00	100.00	100.00		0.01

Field 51 - 11/01/2022: Oats, silage-soft dough

Oats

Sample and source description: Oats

Sample date: 04/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,060.00	1,270.00	18,900.00		9.47
DL	100.00	100.00	100.00		0.01

Field 51 - 06/19/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,950.00	1,240.00	6,460.00		12.00
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

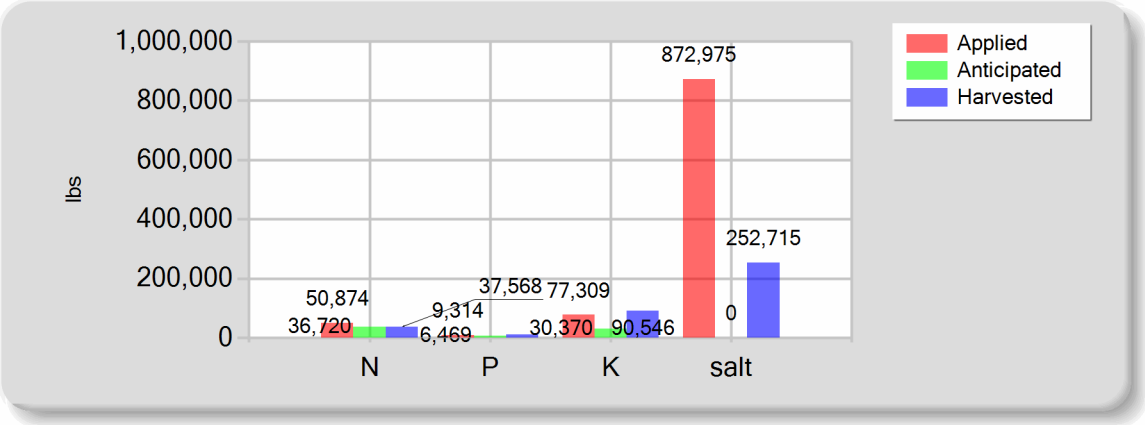
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

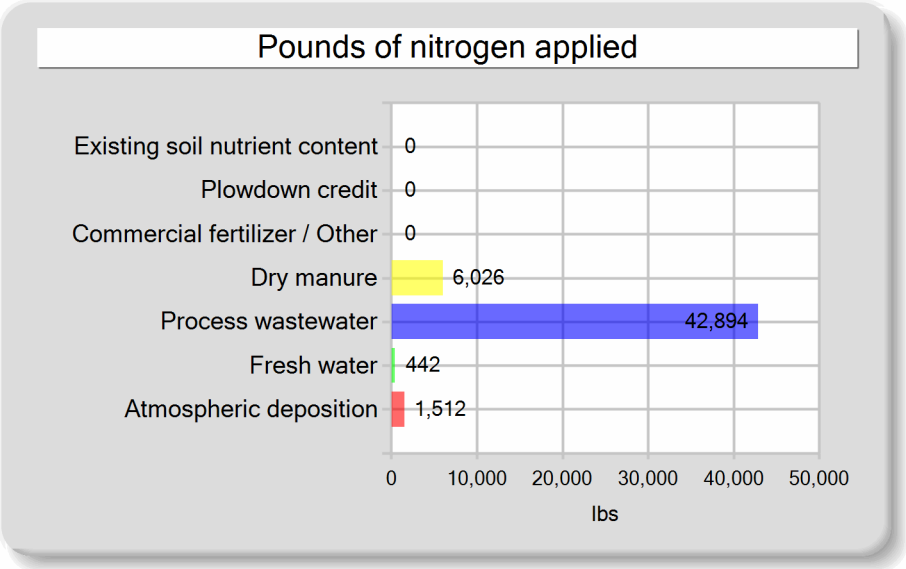
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

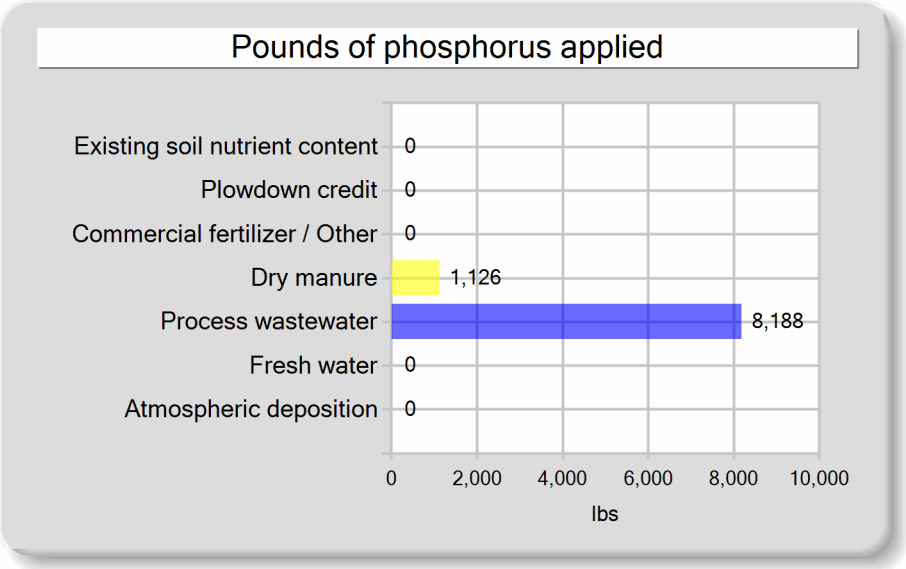
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	6,025.82	1,126.14	2,973.40	158,383.68
Process wastewater	42,893.97	8,188.25	74,335.92	514,278.98
Fresh water	442.16	0.00	0.00	200,312.55
Atmospheric deposition	1,512.00	0.00	0.00	0.00
Total nutrients applied	50,873.95	9,314.39	77,309.32	872,975.21
Anticipated crop nutrient removal	36,720.00	6,469.20	30,369.60	0.00
Actual crop nutrient removal	37,568.46	10,334.46	90,545.90	252,715.29
Nutrient balance	13,305.49	-1,020.07	-13,236.58	620,259.92
Applied to removed ratio	1.35	0.90	0.85	3.45

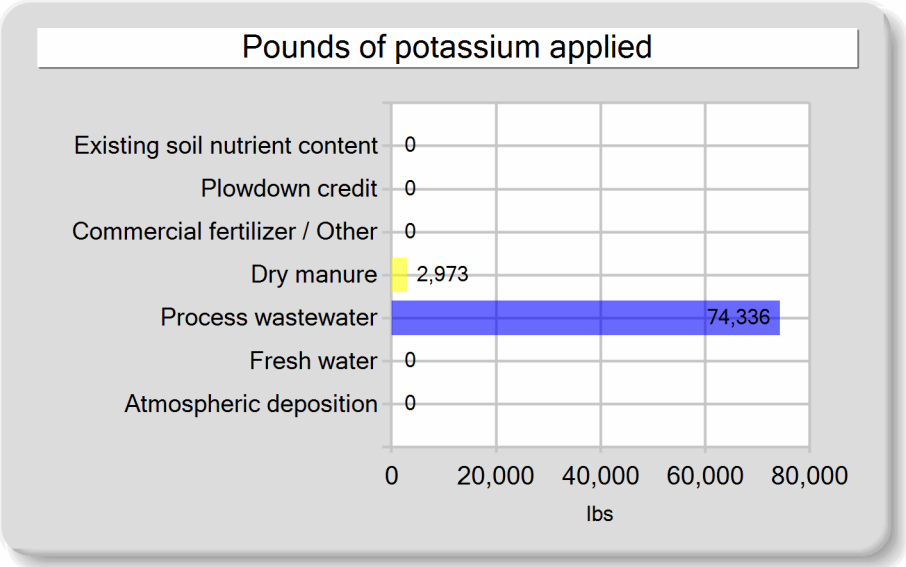
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

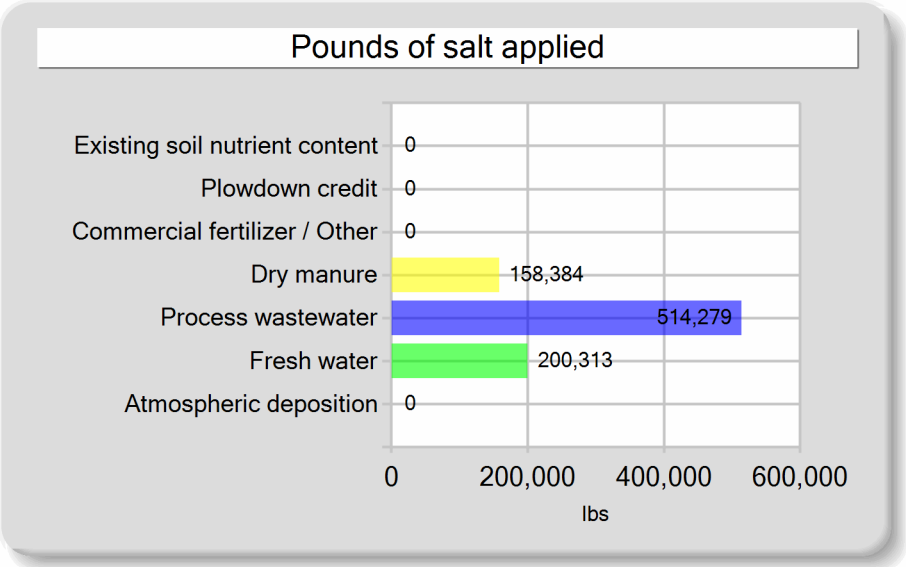


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Disappointing yield on Field 51 forage resulted in nitrogen application to removal ratio exceeding 1.50.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Paul Lopes
Paul Lopes (Apr 18, 2024 23:49 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Paul Lopes	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Apr 18, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
P & D Dairy West
9270 Whitworth Rd.
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/9/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 12.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic	T4003004A	Electrical Conductivity (EC)	EPA 120.1 and	1.42	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	22.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	909	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.8	97.0 (T4003004A)	96.4 (T4003004A)	0.619
Electrical Conductivity (EC)	99.2			1.05
Nitrate Nitrogen (NO3-N)	101	97.9 (T4003004B)	95.9 (T4003004B)	2.04

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
P & D Dairy West
9270 Whitworth Rd.
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/9/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ag Well	T4003004B	Electrical Conductivity (EC)	EPA 120.1 and	1.10	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	3.89	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	505	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	6.69	mg/L
		Soluble Salts (SALT-SOL)	Calculation	706	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	99.8	97.0 (T4003004A)	96.4 (T4003004A)			0.619
Electrical Conductivity (EC)	99.2			1.54 (T4003002B)	1.52 (Dup)	1.05
Nitrate Nitrogen (NO3-N)	101	97.9 (T4003004B)	95.9 (T4003004B)			2.04
Total Dissolved Solids (TDS)	103			435 (T4003007F)	440 (Dup)	1.14
Total Nitrogen (TN)	101			5.60 (T3362069C)	5.60 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.
1232 SOUTH AVENUE
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: P&D Dairy West
Mailing address: 9270 S. Whitworth Road
(not facility address) Gustine CA 95322
Phone#:

Authorized Copy Released To: F&R
Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	1/2	4:00	Domestic		12.0°		X				
2	1/2	4:15	Ag Well		11.2°						H2
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st						
2nd		DENELE ANALYTICAL, INC.	1-3-24 9:10			T4003004

P&D West 2023 AR Complete

Final Audit Report 2024-04-19

Created:	2024-04-18
By:	Joe Ramos (jramos@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAoyeLaJGSmJQ_cxeM3EyxRke29v7pZH0f

"P&D West 2023 AR Complete" History

Document created by Joe Ramos (jramos@fragservices.com)
2024-04-18 - 6:56:58 PM GMT

Document emailed to pandd95@gmail.com for signature
2024-04-18 - 6:57:52 PM GMT

Email viewed by pandd95@gmail.com
2024-04-18 - 9:20:24 PM GMT

Signer pandd95@gmail.com entered name at signing as Paul Lopes
2024-04-19 - 6:49:08 AM GMT

Document e-signed by Paul Lopes (pandd95@gmail.com)
Signature Date: 2024-04-19 - 6:49:10 AM GMT - Time Source: server

Agreement completed.
2024-04-19 - 6:49:10 AM GMT

Adobe Acrobat Sign